

SSIS ETL PROJECT

Technical Use Case & Project Specification

Data Warehouse Population: Union Bank OLTP Banking DWH

Source Database: Union_Bank (OLTP SQL Server)

Target Database: Banking_DWH (Star Schema)

Technology: SQL Server Integration Services (SSIS)

Difficulty Level: Intermediate Advanced

1. Project Overview

This project simulates a real-world ETL (Extract, Transform, Load) scenario in the banking domain. Students will design and implement a complete SSIS solution that migrates transactional data from a normalized OLTP database (Union_Bank) into a dimensional Data Warehouse (Banking_DWH) following the Star Schema model.

The project covers all core SSIS concepts including connection management, data flow components, Slowly Changing Dimensions (SCD), incremental fact loading, and Lookup-based key resolution between dimension and fact tables.

1.1 Databases

Property	Source (OLTP)	Target (DWH)
Database Name	Union_Bank	Banking_DWH
Schema Type	3NF Normalized OLTP	Star Schema (Dimensional)
Server	Local SQL Server Instance	Local SQL Server Instance
Primary Purpose	Day-to-day banking transactions	Analytical reporting & BI

1.2 Learning Objectives

Upon completing this project, the student will be able to:

- Design and configure SSIS packages for ETL workflows
- Implement SCD Type 0, Type 1, and Type 2 patterns for dimension tables
- Build incremental load logic for fact tables using a watermark/timestamp pattern
- Use the Lookup Transformation to resolve surrogate keys between dimensions and the fact table
- Configure OLE DB connections, data type conversions, and error handling
- Organize packages using SSIS Project deployment model and package parameters

2. Source Schema Union_Bank (OLTP)

The source database contains the following tables. Students must analyze the relationships shown in the OLTP ERD before designing their ETL packages.

Table Name	Key Columns	Description
transactions_ATM_Account	TransactionId (PK), TransactionTypeId, AccountNumber, ATM_ID, TransactionDate, Amount	Core fact table recording every ATM transaction
Transcations	Transaction_Type_Id (PK), TransactionType	Lookup table for transaction types (Withdraw, Deposit, Transfer, etc.)
Account	Account_Number (PK), Customer_ID, Account_Type, Balance, Status, Branch_ID	Customer bank accounts with type and balance
Customer	Customer_ID (PK), First Name, Last Name, Email, State, Age, Gender, BD	Core customer demographic table
Card	CardNumber (PK), Card_Type, Card_Status, Expiry_Date, CSV, Account_Number	Debit/credit cards linked to accounts
Atm	ATM_ID (PK), Location, ATM_Status, Branch_ID	ATM machines and their branch location
Branch	BranchId (PK), BranchName	Bank branch master data
Customer_Phones	Mobile_Number (PK), Customer_ID	Multi-valued phone numbers per customer
Loan	LoanID (PK), Amount, Loan_Type, Interest_Rate, Customer_ID, Branch_ID	Loan records per customer
employee	Employee_ID (PK), Branch_ID, Supervisor_ID, Position, Salary	Branch employees
Department	Dnumber (PK), DName	Bank departments

3. Target Schema Banking_DWH (Star Schema)

The target Data Warehouse follows a Star Schema. The central fact table (FactTransaction) is surrounded by four dimension tables. All dimension tables use surrogate primary keys.

3.1 Dimension Tables

3.1.1 Dim accountCustomer (SCD Type 2)

Tracks full historical changes for customer and account attributes.

Column	Data Type	Source Column(s)	Notes
CustomerID	INT IDENTITY	Generated	Surrogate Primary Key
Customer BK	INT	Customer.Customer_ID	Business Key (Natural Key from OLTP)
Accountnumber	VARCHAR	Account.Account_Number	Account business key
AccountType	VARCHAR	Account.Account_Type	SCD Type 1 - overwrite
Balance	DECIMAL	Account.Balance	SCD Type 1 - overwrite
CardNumber	VARCHAR	Card.CardNumber	SCD Type 1 - overwrite
CardType	VARCHAR	Card.Card_Type	SCD Type 1 - overwrite
CardStatus	VARCHAR	Card.Card_Status	SCD Type 1 - overwrite
ExpiryDate	DATE	Card.Expiry_Date	SCD Type 1 - overwrite
fullname	VARCHAR	Customer.First Name + Last Name	Derived: concatenation
state	VARCHAR	Customer.State	SCD Type 2 - track history
age	INT	Customer.Age	SCD Type 1 - overwrite
gender	VARCHAR	Customer.Gender	SCD Type 0 - fixed at insert
DB	DATE	Customer.BD	SCD Type 0 - fixed at insert
StartDate	DATE	System generated	SCD Type 2: record effective date
EndDate	DATE	System generated	SCD Type 2: NULL = current record
AccountStatus	VARCHAR	Account.Status	SCD Type 2 - track history

3.1.2 Dim AtmBranch (SCD Type 1)

ATM and branch details. No history tracking required since ATM location changes simply overwrite the existing record.

Column	Data Type	Source Column(s)	SCD Type
AtmID	INT IDENTITY	Generated	Surrogate PK
AtmBk	INT	Atm.ATM_ID	Business Key

Column	Data Type	Source Column(s)	SCD Type
BranchID	INT	Branch.BranchId	SCD Type 1 - overwrite
BranchName	VARCHAR	Branch.BranchName	SCD Type 1 - overwrite
BranchLocation	VARCHAR	Atm.Location	SCD Type 1 - overwrite

3.1.3 Dim transaction (SCD Type 2)

Transaction type dimension. SCD Type 2 is applied because the TransactionType name might change or be reclassified over time and historical accuracy must be preserved.

Column	Data Type	Source Column(s)	SCD Type
TransactionID	INT IDENTITY	Generated	Surrogate PK
TransactionBK	INT	Transcations.Transaction_Type_Id	Business Key
TransactionType	VARCHAR	Transcations.TransactionType	SCD Type 2 - track history
StartDate	DATE	System generated	SCD Type 2 effective date
EndDate	DATE	System generated	NULL = active record
IsCurrent	BIT	System generated	1 = current, 0 = historical

3.1.4 DimDate (SCD Type 0 Static)

Date dimension is pre-populated once and never updated (Type 0). It is generated independently using a SQL script or SSIS Script Task, covering years 2000-2030.

Column	Notes
DateSK (PK)	Integer surrogate key in YYYYMMDD format (e.g. 20240315)
Date	Full date value
Day, DaySuffix, DayOfWeek, DOWInMonth, DayOfYear	Day-level attributes
WeekOfYear, WeekOfMonth	Week-level attributes
Month, MonthName, Quarter, QuarterName	Month and quarter attributes
Year	4-digit year
StandardDate, HolidayText	Formatted date string and optional holiday flag

3.2 Fact Table FactTransaction (Incremental Load)

Column	Data Type	Source / Derivation	Notes
TransactionID	INT IDENTITY	Generated	Surrogate PK
Transactionbk	INT	transactions_ATM_Account.TransactionId	Business Key

Column	Data Type	Source / Derivation	Notes
TransactionTypeID	INT (FK)	Lookup Dim transaction	Surrogate FK resolved via Lookup
Amount	DECIMAL(18,2)	transactions_ATM_Account.Amount	Transaction amount
TransactionDate	INT (FK)	Lookup DimDate (DateSK)	FK to DimDate
accountNumber	INT (FK)	Lookup Dim accountCustomer	Surrogate FK
AtmId	INT (FK)	Lookup Dim AtmBranch	Surrogate FK
Createdat	DATETIME	GETDATE() at load time	ETL audit column

4. SCD Type Decision Matrix

The following table summarizes the SCD type assigned to every attribute across all dimension tables, along with the business justification.

Dimension	Column	SCD Type	Justification
Dim accountCustomer	gender	Type 0 (Fixed)	Gender does not change; capture once at account opening
Dim accountCustomer	DB (Date of Birth)	Type 0 (Fixed)	Date of birth is immutable
Dim accountCustomer	age, Balance, CardStatus, CardType, CardNumber, ExpiryDate, AccountType	Type 1 (Overwrite)	Current values are sufficient; no historical analysis needed
Dim accountCustomer	state (address), AccountStatus	Type 2 (History)	Customer may relocate or account may be suspended then reactivated; history required
Dim AtmBranch	BranchID, BranchName, BranchLocation	Type 1 (Overwrite)	ATM relocation replaces prior location; no need to track old address
Dim transaction	TransactionType	Type 2 (History)	Transaction type label may be renamed or reclassified; historical transactions must preserve original classification
DimDate	All columns	Type 0 (Fixed)	Date dimension is static; calendar facts never change

5. ETL Architecture & Package Design

5.1 Solution Structure

The SSIS solution must be organized as a Project Deployment Model with the following package structure:

Package Name	Type	Description
Master.dtsx	Control Flow	Parent package that executes all child packages in the correct dependency order using Execute Package Tasks
Load_DimDate.dtsx	One-time / Annual	Populates DimDate using a Script Task or SQL Task. Runs once and only when the table is empty.
Load_DimAccountCustomer.dtsx	SCD Type 1 & 2	Loads customer + account + card data with mixed SCD handling
Load_DimAtmBranch.dtsx	SCD Type 1	Loads ATM and Branch data; overwrites changed attributes
Load_DimTransaction.dtsx	SCD Type 2	Loads transaction types with full history tracking
Load_FactTransaction.dtsx	Incremental	Loads new ATM transactions since last run using watermark timestamp

5.2 Package Execution Order (Master Package)

The Master package must enforce the following dependency sequence to ensure all dimension surrogate keys exist before the fact table load begins:

1. Load_DimDate.dtsx (prerequisite, run if empty)
2. Load_DimAccountCustomer.dtsx
3. Load_DimAtmBranch.dtsx
4. Load_DimTransaction.dtsx
5. Load_FactTransaction.dtsx (runs only after all dims succeed)

Use Precedence Constraints set to Success between each Execute Package Task to enforce this order.

5.3 Project-Level Connection Managers

Define the following shared connection managers at the project level so all packages can reuse them:

Connection Manager Name	Type	Points To
CM_OLTP	OLE DB	Union_Bank (source SQL Server database)
CM_DWH	OLE DB	Banking_DWH (target SQL Server database)

5.4 Project Parameters

Parameter Name	Data Type	Default Value	Usage
OLTP_Server	String	localhost	Source server name
DWH_Server	String	localhost	Target server name
BatchSize	Int32	10000	Rows per data flow buffer

6. Detailed Package Specifications

6.1 Load_DimDate.dtsx

Control Flow

- Execute SQL Task: Check IF COUNT(*) FROM DimDate = 0
- If true execute Script Task or SQL Script to generate date rows from 2000-01-01 to 2030-12-31
- If false skip (using Precedence Constraint with Expression)

Key Requirement

DateSK must be an integer in YYYYMMDD format (e.g., 20240315) to allow direct integer join with TransactionDate in FactTransaction.

6.2 Load_DimAccountCustomer.dtsx (SCD Mixed: Type 0, 1, 2)

Source Query (OLE DB Source CM_OLTP)

Join the following OLTP tables to produce a flat staging dataset:

- Customer Account (JOIN on Customer_ID)
- Account Card (JOIN on Account_Number)

Derive fullname as LTRIM(RTRIM([First Name])) + ' ' + LTRIM(RTRIM([Last Name]))

Data Flow Steps

6. OLE DB Source: query joins Customer + Account + Card
7. Derived Column: compute fullname
8. Lookup Transformation: match on Accountnumber against Dim accountCustomer
 - Match found route to SCD Type 1 update branch (overwrite Balance, CardStatus, age, etc.)
 - No match route to OLE DB Destination (INSERT new row with StartDate=today, EndDate=NULL, IsCurrent is not applicable here)
9. For Type 2 columns (state, AccountStatus): if changed value detected, expire old row (UPDATE EndDate=today-1) and INSERT new row with new StartDate
10. OLE DB Command: UPDATE existing rows for Type 1 changes
11. OLE DB Destination: INSERT new rows

SCD Type 2 Implementation Pattern

Because SSIS's built-in SCD Wizard generates slow performance for large tables, students must implement SCD Type 2 manually using the following pattern:

12. Lookup on business key (Customer BK + Accountnumber)
13. Conditional Split: compare incoming state/AccountStatus with existing values
14. OLE DB Command: UPDATE old row SET EndDate = GETDATE()-1 WHERE EndDate IS NULL AND CustomerID = ?
15. OLE DB Destination: INSERT new row with StartDate = GETDATE(), EndDate = NULL

6.3 Load_DimAtmBranch.dtsx (SCD Type 1)

Source Query (CM_OLTP)

```
SELECT a.ATM_ID, a.Location, a.Branch_ID, b.BranchName FROM Atm a JOIN Branch b ON a.Branch_ID = b.BranchId
```

Data Flow Steps

16. OLE DB Source: above query
17. Lookup Transformation: match AtmBk against [Dim AtmBranch].AtmBk in CM_DWH
 - No Match (new ATM) OLE DB Destination INSERT
 - Match Found Conditional Split check if BranchName or Location changed
 - Changed OLE DB Command UPDATE BranchID, BranchName, BranchLocation
 - No Change Ignore (discard row)

6.4 Load_DimTransaction.dtsx (SCD Type 2)

Source Query (CM_OLTP)

```
SELECT Transaction_Type_Id, TransactionType FROM Transcations
```

Data Flow Steps

18. OLE DB Source
19. Lookup: match TransactionBK against [Dim transaction].TransactionBK
 - No Match INSERT new row (StartDate=today, EndDate=NULL, IsCurrent=1)
 - Match + TransactionType changed expire old row, INSERT new row
 - Match + no change discard

6.5 Load_FactTransaction.dtsx (Incremental Load)

Watermark Strategy

A control table is used to track the last successfully loaded transaction date. This pattern ensures only new rows are extracted on each run.

Step	SSIS Component	Description
1	Execute SQL Task	Read LastLoadDate from meta_Control_Fact: SELECT LastLoadDate FROM meta_Control_Fact WHERE TableName = 'FactTransaction'
2	Variable Assignment	Store result in User::LastLoadDate (DateTime variable)
3	OLE DB Source	Extract new rows: SELECT * FROM transactions_ATM_Account WHERE TransactionDate > ? (map LastLoadDate parameter)
4	Lookup Dim transaction	Match TransactionTypeld (OLTP BK) to TransactionID (DWH SK) on IsCurrent=1

Step	SSIS Component	Description
5	Lookup Dim accountCustomer	Match Account_Number (OLTP BK) to CustomerID (DWH SK) where EndDate IS NULL
6	Lookup Dim AtmBranch	Match ATM_ID (OLTP BK) to AtmID (DWH SK)
7	Derived Column	Convert TransactionDate to integer DateSK: YEAR * 10000 + MONTH * 100 + DAY
8	Lookup DimDate	Match computed DateSK integer to DimDate.DateSK
9	OLE DB Destination	INSERT resolved rows into FactTransaction
10	Execute SQL Task	UPDATE meta_Control_Fact SET LastLoadDate = GETDATE() WHERE TableName = 'FactTransaction'

Lookup Error Handling

For all Lookup Transformations in the fact package:

- Redirect no-match rows to a Flat File Destination (error log file) rather than failing the package
- Add a Derived Column to tag rejected rows with a rejection reason before writing to the error file
- Log the count of rejected rows to meta_Control_Fact or a separate audit table

8. Project Deliverables

The student must submit the following:

#	Deliverable	Description
1	SSIS Solution (.sln)	Full Visual Studio SSIS project with all packages organized under a project deployment model
2	Dimension Packages (x3)	One package per dimension with correct SCD type implementation as specified
3	Fact Package	Load_FactTransaction.dtsx with watermark-based incremental load and all four Lookup transformations
4	SQL Scripts	Scripts to: (a) meta_data
5	Test Evidence	Screenshots showing: row counts before/after, SCD Type 2 history records, incremental load proof (second run loads 0 new rows if no new data)

9. Implementation Tips & Common Mistakes

Common Mistakes to Avoid

- Never truncate dimension tables between runs - this destroys surrogate key integrity with the fact table
- Do not use the built-in SCD Wizard for Type 2 on large tables - implement manually for performance
- Always join Lookup on the business key (BK), never on the surrogate key when resolving OLTP to DWH

Best Practices

- Use OLE DB Command (not OLE DB Destination) for UPDATE operations on existing dimension rows
- Add Data Viewers during development to inspect row counts at each transformation output
- Set Lookup cache mode to Full Cache for dimension tables to improve performance
- Use package-level variables to pass LastLoadDate from the Execute SQL Task to the OLE DB Source
- Test the second run immediately after the first to confirm 0 new rows are loaded (idempotency test)